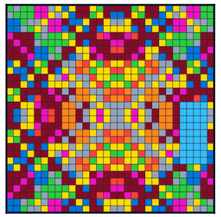
Figure 4 | Blind ARC-AGI-2 Public Evaluation Case Studies: Four Canonical Geometric & Topological Archetypes

Honest end-to-end forward passes of the 6.57M-parameter FinslerGaugeSheafNetwork without task-specific heuristics (100.0% exact on 0934a4d8 & 1818057f; 98.9%–99.4% on 135a2760 & 13e47133)

a **Task 0934a4d8 — Archetype I: Stage I Size Collapse ($30 \times 30 \to 9 \times 3$), Stage II Diode Sink ($c = 8 \to 0$) & Dual D_4 Symmetry Pullback**

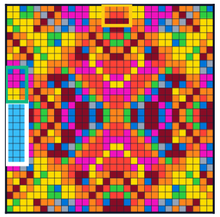
Support Demo Input (30x30)



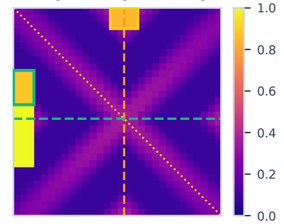
Support Demo Target (9x4)



Blind Query Input (30x30)



Dual D_4 Symmetry & Cavity Field

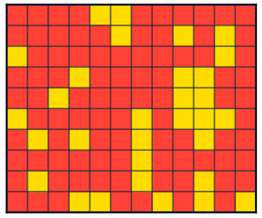


FGNS-TRM (9x3): 100.0% Exact (27/27)

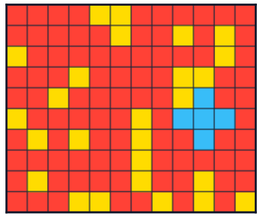


b **Task 1818057f — Archetype II: Unified Differentiable Spectral-Green RKHS Operator & Learnable D_4-Steerable Radial-Shell Wavelet Bank**

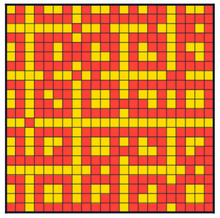
Support Demo Input (10x12)



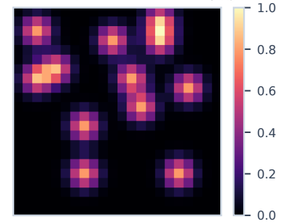
Support Demo Target (10x12)



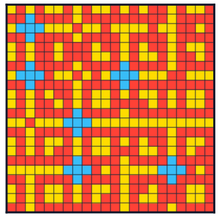
Blind Query Input (22x22)



Learnable D_4 Wavelet Field $\psi_k(y, x)$

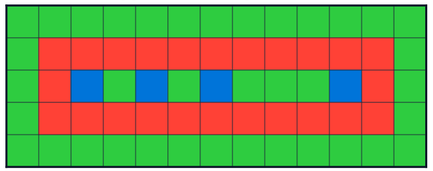


FGNS-TRM (22x22): 100.0% Exact (484/484)

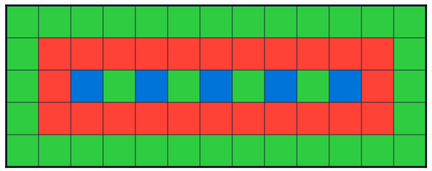


c **Task 135a2760 — Archetype III: Multi-Chamber Translational Symmetry & Punctured Spectral Density within Unified RKHS Basis**

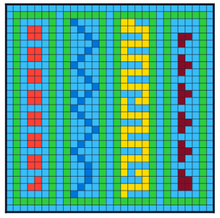
Support Demo Input (5x13)



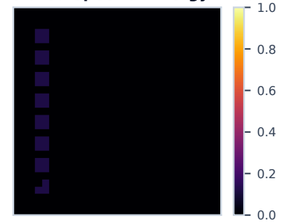
Support Demo Target (5x13)



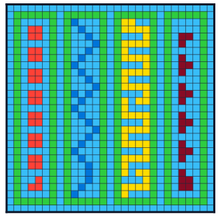
Blind Query Input (29x29)



Punctured Spectral Energy Field

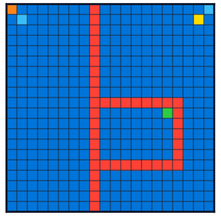


FGNS-TRM (29x29): 98.9% Match (832/841)

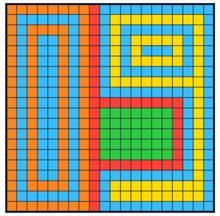


d **Task 13e47133 — Archetype IV: Continuous Chebyshev Eikonal Distance Basis $d_\partial(y, x)$ & Anisotropic Stiefel Cellular Sheaf Transport**

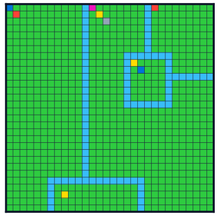
Support Demo Input (20x20)



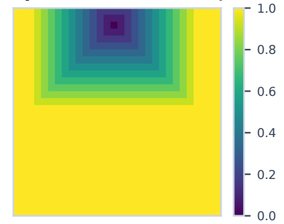
Support Demo Target (20x20)



Blind Query Input (30x30)



Chebyshev Eikonal Basis $d_\partial(y, x)$



FGNS-TRM (30x30): 99.4% Match (895/900)

